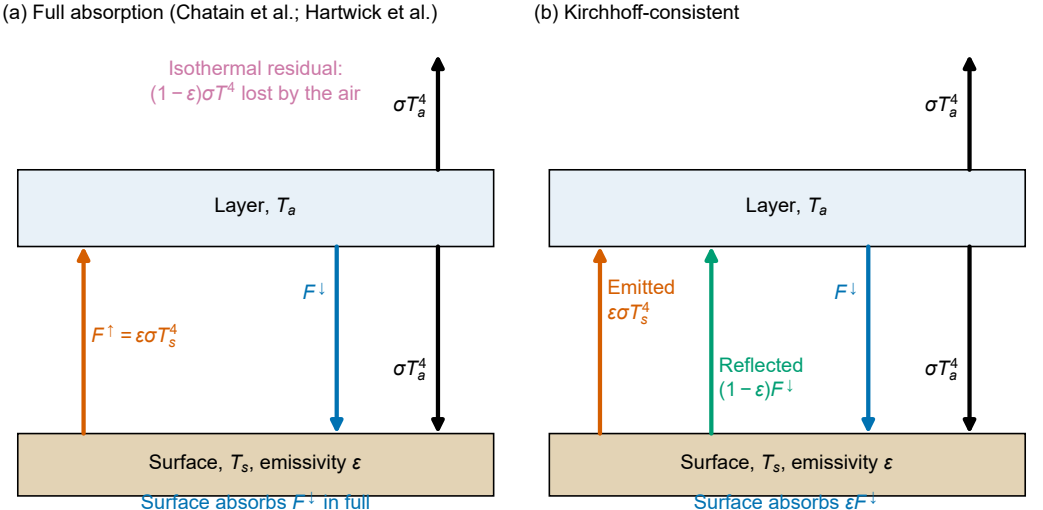
(a) Full absorption (Chatain et al.; Hartwick et al.)
Isothermal residual:
$(1-\varepsilon)\sigma T^4$ lost by the air
σT_a^4
Layer, T_a
$F^{\uparrow} = \varepsilon\sigma T_s^4$
$F^{\downarrow}$
σT_a^4
Surface, T_s, emissivity ε
Surface absorbs $F^{\downarrow}$ in full
(b) Kirchhoff-consistent
σT_a^4
Layer, T_a
Emitted
$\varepsilon\sigma T_s^4$
Reflected
$(1-\varepsilon)F^{\downarrow}$
$F^{\downarrow}$
σT_a^4
Surface, T_s, emissivity ε
Surface absorbs $\varepsilon F^{\downarrow}$